

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMLLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Content-level validity is the dimension where ArabicMMLU diverges most sharply from the deployment's eight-country requirement. While the benchmark's content is sourced from real native Arabic exams **[Q1, Q26]** — a clear strength over translated alternatives **[Q5, Q6]** — the empirical content distribution is heavily skewed: civics, social science, history, and economics at school level are dominated by Jordanian-specific institutional, royal, and constitutional content presented as universal curriculum knowledge **[DATASET-D1, D2, D3, D4, D5, D6, D7, D20, D21, D22, D25, D30, D31]**. University-level economics, accounting, management, and political science come exclusively from a single Egyptian university (Assiut University) in the sample **[DATASET-D13, D14, D28, D39, D40]**, and Law (Professional) is exclusively Moroccan in the observed sample **[DATASET-D8, D9, D10, D34]**. The deployment user explicitly requires country-specific tailoring for law [elicitation A2] and curriculum-aligned content for each of the eight countries [elicitation A1]. Kuwait has no observed examples and no documented annotators **[Q31, Q96]**; Morocco has no documented annotators **[Q31, Q96]**. Egypt-specific currency (ج.م.ع) appears in university economics presented as universal **[DATASET-D28]**. The benchmark's own admission that countries are represented unequally **[Q89]** is empirically confirmed and amplified by the dataset analysis: rather than uniform under-coverage, the data shows Jordanian-curriculum content is propagated as default for non-Jordanian cohorts. This generates substantial construct-irrelevant variance for Moroccan, Emirati, Kuwaiti, Saudi, Lebanese, and Palestinian students.

ID	Evidence
Q1	"we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions." (p.1)
Q26	"The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA)." (p.3)
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D13	أي عيب تلاحظ في نموذج الحاسب الآلي؟ — مفهوم التدقيق الحاسوبي — general accounting principle, from Egyptian source
D8	أي عيب تلاحظ في نموذج دعوى الاستئناف؟ — قواعد الدعاوى المدنية في القانون المغربي — Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D28	العملة المصرية هي... — ١٠٠٠٠ ج.م.ع — Egyptian currency (ج.م.ع = Egyptian pound) in university economics — Egypt-specific monetary context

D2	— "The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D3	with answer "The council with general sovereignty over all state affairs is the Council of Ministers" — specifically Jordanian constitutional structure
D4	— "Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D5	"The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" with options referring to Jordanian kings — Jordanian royal history
D6	www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D20	Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services
D21	King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D22	"Lady of the Mountain Church is in Anjara [Jordan]" — Jordanian religious geography
D25	The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history
D30	Wadi Rum is a Jordanian nature reserve under joint management" — Jordan-specific geography fact
D31	King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
D9	عافدلاب ةفلكملم ةيموكحل ةطلسل او لدعل ربزو نم راربب...نيكيكردلل ةيئاضقلل ةطرشلل طباض ةقص لي وخت نكمي — Moroccan judicial police powers — references "يكلمل كرل" (Royal Gendarmerie of Morocco)
D10	Moroccan civil party withdrawal in criminal proceedings
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF

Strengths

- Native exam provenance avoids translation artifacts that plague competing Arabic benchmarks [Q5, Q6]
- STEM content (math, biology, physics, CS factual) is genuinely curriculum-neutral and reliable across all eight countries [DATASET-D24]
- Palestine is substantively represented in STEM subjects (Biology HS, CS Primary, Math Primary) [DATASET-D23, D24]
- Some Moroccan legal content is present, even if narrow [DATASET-D8, D9, D10]
- Native annotators and exam URLs document content provenance per question [Q32, Q33]

ID	Evidence
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)

D28	Egyptian currency (جني = ١٠٠٠٠٠ Egyptian pound) in university economics — Egypt-specific monetary context
D8	— يوعدل رهوج يف مكحل رودص دعب الإ عوفد وأ ضراع عازن يف ةرداصل وأ أديهمم المتل ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D2	— "The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D4	— "Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D5	— "The 1952 constitution was issued during the reign of King [Talal bin Abdullah]" — Jordanian royal history
D6	— "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo" — explicitly references Jordan's central bank
D20	— "Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services
D21	— "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D25	— "The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history
D31	— King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history
D39	Source: http://www.aun.edu.eg/commerce/ ... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
D9	— عافدللاب ةفل كملم ةيموكحل ةطلسلال ولدعل ربزو نم رارقب...نيكي كيدلل ةيئاضقلل ةطرشلل طباض ةقص لي وخت نكمي Moroccan judicial police powers — references "يكللم كرذل" (Royal Gendarmerie of Morocco)
D10	— Moroccan civil party withdrawal in criminal proceedings
D17	— "ركذ ام عي مج: ةدع ةي باجي راثأ ري كفتلل" — Country: Jordan — 5-option Islamic studies question about positive effects of thinking
D37	— "ركذ ام عي مج: مي ركلل أنرقل ةوالل بآأ نم" — Country: Jordan — Generic Islamic etiquette of Quran recitation — no madhab specificity indicated
D38	— "ركذ ام عي مج: مي ركلل أنرقل يف لاثمأل داري ةئواف" — Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
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Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF

Information Gaps

- Whether Jordanian dominance in civics/history/social science extends to the full benchmark or is concentrated in specific level/subject configurations [gap_id 5]
- Whether Law subset beyond the sampled Moroccan questions includes other jurisdictions [gap_id 4]
- Madhab-specific framing in Islamic studies is undocumented [gap_id 7]

Requires Expert Verification

- Per-country curriculum review by national-curriculum experts from Kuwait, Morocco, Palestine, UAE to identify questions presented as 'universal' that are actually national-curriculum-specific
- Islamic studies curriculum experts to assess madhab calibration risk for Moroccan Maliki and Saudi Hanbali contexts

Remediation

Gap 1: Jordanian-curriculum civics, history, social science, and high-school economics content is encoded as universal across configurations, creating construct-irrelevant variance for the seven non-Jordanian deployment cohorts [DATASET-D1, D2, D6, D7, D21, D25].

Recommendation: Use the per-question country metadata field [Q33] to filter or stratify evaluation by source country; do not aggregate across countries for civics, history, or social science. Build a Kuwaiti, Moroccan, and Palestinian curriculum supplement before deploying to those cohorts.

Gap 2: Islamic studies has no madhab labeling anywhere in the benchmark [gap_id 7]; observed Islamic Studies HS content is Jordanian-curriculum-sourced [DATASET-D38] without Maliki (Morocco) or Hanbali (KSA) calibration.

Recommendation: Engage Islamic studies curriculum experts from Morocco (Maliki), KSA (Hanbali), and a Shafi'i-region representative to audit a stratified sample of Islamic studies items and tag madhab-specific items; flag madhab-divergent questions for country-conditioned evaluation.

ID	Evidence
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D2	The first coalition government in Jordan in 1952 was known as the ___ government — Jordanian political history
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D7	0.67 — 67% of Jordanians cannot access formal financial services — Jordan-specific statistic as universal economic fact
D21	King Talal bin Hussein's reign ended for health reasons in 1952 — one year — Jordanian royal history in primary school
D25	The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions — Jordanian national history
Q33	In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key. (p.4)
D38	Country: Jordan — Islamic studies question about examples in Quran — Jordanian curriculum framing